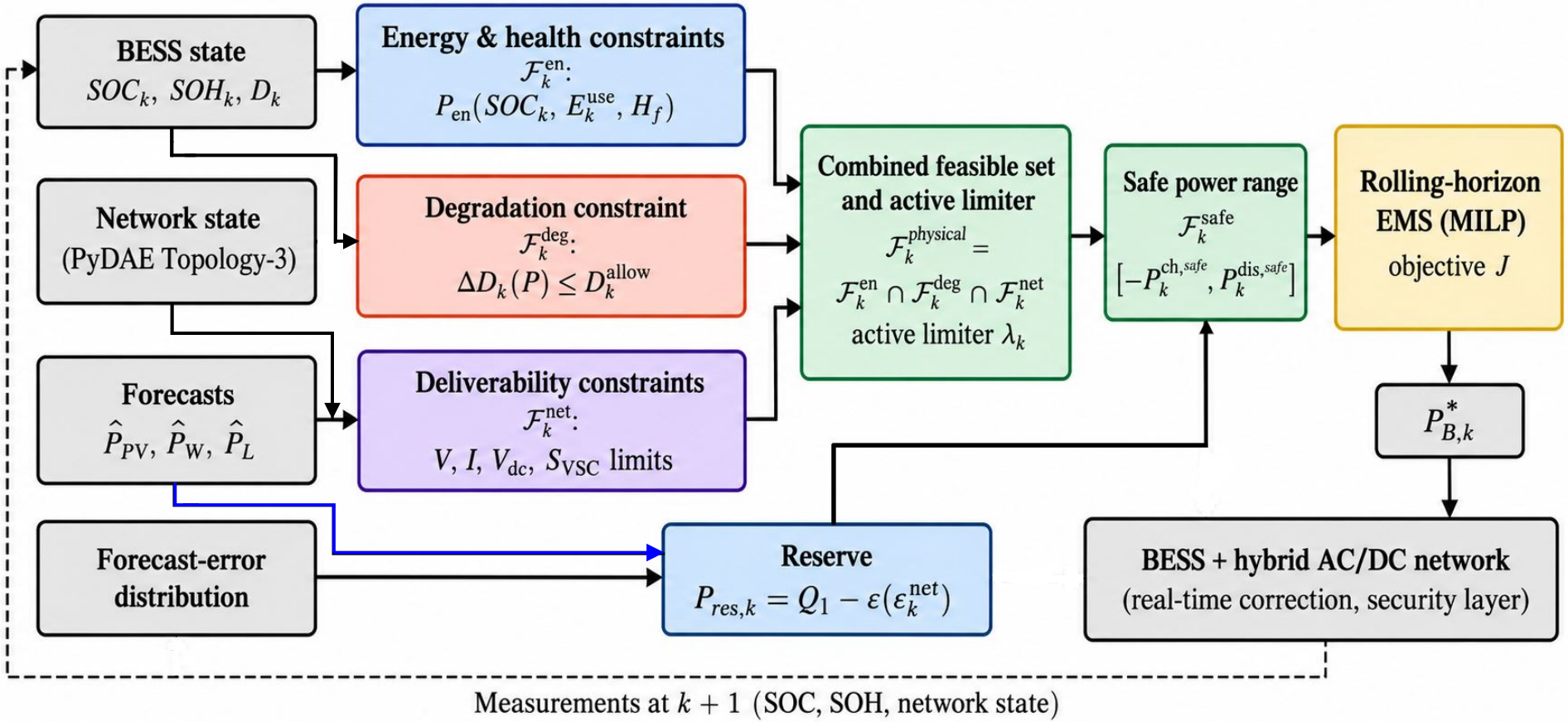

BESS state
SOC_k, SOH_k, D_k
Energy & health constraints
$\mathcal{F}_k^{\text{en}}$:
$P_{\text{en}}(SOC_k, E_k^{\text{use}}, H_f)$
Network state
(PyDAE Topology-3)
Degradation constraint
$\mathcal{F}_k^{\text{deg}}$:
$\Delta D_k(P) \le D_k^{\text{allow}}$
Forecasts
$\hat{P}_{\text{PV}}$, $\hat{P}_{\text{W}}$, $\hat{P}_L$
Deliverability constraints
$\mathcal{F}_k^{\text{net}}$:
V, I, V_{dc}, S_{VSC} limits
Forecast-error
distribution
Reserve
$P_{res,k} = Q_1 - \varepsilon(\varepsilon_k^{\text{net}})$
Combined feasible set
and active limiter
$\mathcal{F}_k^{\text{physical}} =$
$\mathcal{F}_k^{\text{en}} \cap \mathcal{F}_k^{\text{deg}} \cap \mathcal{F}_k^{\text{net}}$
active limiter λ_k
Safe power range
$\mathcal{F}_k^{\text{safe}}$
$[-P_k^{\text{ch,safe}}, P_k^{\text{dis,safe}}]$
Rolling-horizon
EMS (MILP)
objective J
$P_{B,k}^*$
BESS + hybrid AC/DC network
(real-time correction, security layer)
Measurements at $k+1$ (SOC, SOH, network state)